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Unit – 4 ⇝ Advanced Matrix Algebra (Tutorial Manual)

Method 1 ⇝ Eigenvalues and Eigenvectors

Examples of Method-1: Eigenvalues and Eigenvectors

B	1	Find the eigenvalues of the following matrices: $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 3 \\ 1 & 3 \end{bmatrix}$, $C = \begin{bmatrix} 10 & -9 \\ 4 & -2 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix}$, $E \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$, $F = \begin{bmatrix} 3 & 0 & -5 \\ \frac{1}{5} & -1 & 0 \\ 1 & 1 & -2 \end{bmatrix}$. **Answer: A ⇝ 1, 2, B ⇝ 2, 6, C ⇝ 4, 4, D ⇝ 1, 2, 2,** **E ⇝ 1, 2, 3, F ⇝ 0, $-\sqrt{2}$, $\sqrt{2}$**
A	2	Find the eigenvalues and eigenvectors of the matrix $A = \begin{bmatrix} 1 & -2 \\ -5 & 4 \end{bmatrix}$. **Answer: $\lambda = -1$ ⇝ $X = [1 \quad 1]^T$, $\lambda = 6$ ⇝ $X = \left[-\frac{2}{5} \quad 1\right]^T$**
A	3	Find the eigenvalues and eigenvectors of the matrix $A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$. **Answer: $\lambda = 2$ ⇝ $X = [-1 \quad 1 \quad 0]^T$, $\lambda = 3$ ⇝ $X = [1 \quad 0 \quad 0]^T$** **$\lambda = 5$ ⇝ $X = [3 \quad 2 \quad 1]^T$**
A	4	Find the eigenvalues and corresponding eigenvectors of the matrix $A = \begin{bmatrix} 4 & 1 & 2 \\ 0 & 3 & -1 \\ 0 & 0 & 2 \end{bmatrix}$. **Answer: $\lambda = 4$ ⇝ $X = [1 \quad 0 \quad 0]^T$, $\lambda = 3$ ⇝ $X = [-1 \quad 1 \quad 0]^T$** **$\lambda = 2$ ⇝ $X = [-1.5 \quad 1 \quad 1]^T$**

C	5	Find the eigenvalues and eigenvectors of the matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -3 & 3 \end{bmatrix}$. **Answer:** $\lambda \rightarrowtail \mathbf{1},\ \mathbf{1},\ \mathbf{1}, \qquad \mathbf{X} = [\,\mathbf{1}\quad \mathbf{1}\quad \mathbf{1}\,]^{\mathbf{T}}$
B	6	Find the eigenvalues and corresponding eigenvectors for the matrix $A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$. **Answer:** $\lambda = \mathbf{1},\ \mathbf{2},\ \mathbf{2}, \qquad \mathbf{X} = [\,-\mathbf{1}\quad -\mathbf{1}\quad \mathbf{1}\,]^{\mathbf{T}},\ [\,\mathbf{2}\quad \mathbf{1}\quad \mathbf{0}\,]^{\mathbf{T}}$
A	7	Find the eigenvalues and eigen space for the eigen space of the matrix $A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$. **Answer:** $\lambda = \mathbf{1}, \qquad \mathbf{2}$ $\mathbf{E_1} = \left\{ \begin{bmatrix} \mathbf{1} \\ \mathbf{0} \end{bmatrix} \cdot \mathbf{t} \mid \mathbf{t} \in \mathbb{R} \right\}, \qquad \mathbf{E_2} = \left\{ \begin{bmatrix} \mathbf{1} \\ \mathbf{1} \end{bmatrix} \cdot \mathbf{t} \mid \mathbf{t} \in \mathbb{R} \right\}$
B	8	Find the eigenvalues and eigen space for the eigen space of the matrix $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. **Answer:** $\lambda = -\mathbf{1},\ -\mathbf{1},\ \mathbf{2}$ $\mathbf{E_{-1}} = \left\{ \begin{bmatrix} -\mathbf{1} \\ \mathbf{1} \\ \mathbf{0} \end{bmatrix} \cdot \mathbf{t_1} + \begin{bmatrix} -\mathbf{1} \\ \mathbf{0} \\ \mathbf{1} \end{bmatrix} \cdot \mathbf{t_2} \mid \mathbf{t_1}, \mathbf{t_2} \in \mathbb{R} \right\},$ $\mathbf{E_2} = \left\{ \begin{bmatrix} \mathbf{1} \\ \mathbf{1} \\ \mathbf{1} \end{bmatrix} \cdot \mathbf{t} \mid \mathbf{t} \in \mathbb{R} \right\}$

B	9	Let $A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$, then find eigenvalues, eigenvectors corresponding to each eigenvalue of A and write eigen space for each eigenvalue. **Answer:** $\lambda = 1 \rightsquigarrow X = [-1 \quad 1 \quad 0]^T$, $\lambda = 2 \rightsquigarrow X = \left[-1 \quad \frac{1}{2} \quad 1\right]^T$ $\lambda = 3 \rightsquigarrow X = \left[-\frac{1}{2} \quad \frac{1}{2} \quad 1\right]^T$, $E_1 = \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \cdot t \;\middle|\; t \in \mathbb{R} \right\}$ $E_2 = \left\{ \begin{bmatrix} -1 \\ \frac{1}{2} \\ 1 \end{bmatrix} \cdot t \;\middle|\; t \in \mathbb{R} \right\}$, $E_3 = \left\{ \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ 1 \end{bmatrix} \cdot t \;\middle|\; t \in \mathbb{R} \right\}$
A	10	Find the eigenvalues of $A = \begin{bmatrix} 0 & 3 & 2 \\ -2 & 5 & 2 \\ -3 & 3 & 5 \end{bmatrix}$. Is it invertible? **Answer:** 2, 3, 5 & Yes
B	11	If $A = \begin{bmatrix} 13 & 18 & -16 \\ -14 & -9 & 11 \\ -10 & 0 & 4 \end{bmatrix}$, then find the eigenvalues of the matrix A^T. Is A invertible? **Answer:** $\lambda = -1, 0, 9$, No
B	12	Let $A = \begin{bmatrix} 1 & 0 \\ 8 & -2 \end{bmatrix}$, then find eigenvalues of A, A^{10}, 4A, A^{-3}, A^T, $A + I$ & $A^2 - 3I$. **Answer:** $A \rightsquigarrow 1, -2$; $A^{10} \rightsquigarrow 1, 1024$; $4A \rightsquigarrow 4, -8$ $A^{-3} \rightsquigarrow 1, -\frac{1}{8}$; $A^T \rightsquigarrow 1 \; -2$; $A + I \rightsquigarrow 2, -1$ $A^2 - 3I \rightsquigarrow -2, 1$

A	13	If $A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix}$, then find the eigenvalues of A^T and 5A. **Answer: A^T ⇝ 1, 2, 2, 5A ⇝ 5, 10, 10**
A	14	Find the eigenvalues of A^9 for $A = \begin{bmatrix} 1 & 3 & 7 & 11 \\ 0 & 0.5 & 3 & 8 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 2 \end{bmatrix}$. **Answer: A^9 ⇝ 1, $(0.5)^9$, 0, 2^9**
A	15	Find the eigenvalues of $A = \begin{bmatrix} 1 & 0 & 0 \\ 5 & 2 & 0 \\ 12 & 15 & 3 \end{bmatrix}$ and hence find the eigenvalues of A^5 and A^{-1}. **Answer: A ⇝ 1, 2, 3, A^5 ⇝ 1, 2^5, 3^5, A^{-1} ⇝ 1, $\frac{1}{2}$, $\frac{1}{3}$**

Method 2 ⇝ Cayley – Hamilton Theorem

Examples of Method-2: Cayley – Hamilton Theorem

A	1	Verify Cayley – Hamilton theorem for $A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$ and hence find A^{-1}. **Answer:** $\frac{1}{5}\begin{bmatrix} 3 & 1 \\ -2 & 1 \end{bmatrix}$
A	2	Verify Cayley – Hamilton theorem for $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ and hence find A^{-2}. **Answer:** $\frac{1}{9}\begin{bmatrix} 5 & -4 \\ -4 & 5 \end{bmatrix}$
A	3	Using Cayley – Hamilton theorem, find A^{-1} and A^{-2} for $A = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$. **Answer:** $A^{-1} = \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$, $A^{-2} = \begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix}$
B	4	Verify Cayley – Hamilton theorem and use it to find the inverse of $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix}$. **Answer:** $\begin{bmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{bmatrix}$
B	5	Using Cayley – Hamilton theorem, find the inverse of $A = \begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix}$. **Answer:** $\frac{1}{35}\begin{bmatrix} -4 & 11 & -5 \\ -1 & -6 & 25 \\ 6 & 1 & -10 \end{bmatrix}$

B	6	State Cayley – Hamilton theorem and verify for the matrix, also find the inverse of matrix A. $A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$. **Answer:** $\frac{1}{6}\begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{-1} \\ \mathbf{2} & \mathbf{6} & \mathbf{-2} \\ \mathbf{2} & \mathbf{0} & \mathbf{4} \end{bmatrix}$
B	7	Verify Cayley – Hamilton theorem for $A = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \\ 2 & 3 & 1 \end{bmatrix}$ and hence find A^4. **Answer:** $\begin{bmatrix} \mathbf{193} & \mathbf{160} & \mathbf{144} \\ \mathbf{224} & \mathbf{177} & \mathbf{160} \\ \mathbf{272} & \mathbf{224} & \mathbf{193} \end{bmatrix}$
A	8	Verify Cayley – Hamilton theorem for $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ and hence find A^4. **Answer:** $\begin{bmatrix} \mathbf{41} & \mathbf{40} \\ \mathbf{40} & \mathbf{41} \end{bmatrix}$
C	9	Verify Cayley – Hamilton theorem for $A = \begin{bmatrix} 6 & -1 & 1 \\ -2 & 5 & -1 \\ 2 & 1 & 7 \end{bmatrix}$ and hence find A^4. **Answer:** $\begin{bmatrix} \mathbf{2176} & \mathbf{-520} & \mathbf{1400} \\ \mathbf{-1920} & \mathbf{776} & \mathbf{-1400} \\ \mathbf{1920} & \mathbf{520} & \mathbf{2696} \end{bmatrix}$

C	10	Verify Cayley – Hamilton theorem for matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and hence find the matrix represented by $A^6 - 4A^5 + 3A^4 + 2A^3 + 2A^2 + 3A + I$. **Answer:** $\begin{bmatrix} \mathbf{45} & \mathbf{37} & \mathbf{37} \\ \mathbf{0} & \mathbf{8} & \mathbf{0} \\ \mathbf{37} & \mathbf{37} & \mathbf{45} \end{bmatrix}$
C	11	Verify Cayley – Hamilton theorem for matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and hence find the matrix represented by $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$. **Answer:** $\begin{bmatrix} \mathbf{8} & \mathbf{5} & \mathbf{5} \\ \mathbf{0} & \mathbf{3} & \mathbf{0} \\ \mathbf{5} & \mathbf{5} & \mathbf{8} \end{bmatrix}$

Method 3 ⇝ Algebraic Multiplicity and Geometric Multiplicity

Examples of Method-3: Algebraic Multiplicity and Geometric Multiplicity

A	1	Find AM and GM for each of the eigenvalue of matrix $A = \begin{bmatrix} 4 & -2 & 0 & 7 \\ 0 & 3 & 1 & -2 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$. **Answer: For $\lambda = 2$: AM $= 2$ and GM $= 2$** **For $\lambda = 3$: AM $= 1$ and GM $= 1$** **For $\lambda = 4$: AM $= 1$ and GM $= 1$**
B	2	Find dimension of eigenspace for each eigenvalues of matrix $A = \begin{bmatrix} -1 & 4 & -2 \\ -3 & 4 & 0 \\ -3 & 1 & 3 \end{bmatrix}$. **Answer: For $\lambda = 1$: Dimension of eigenspace $=1$** **For $\lambda = 2$: Dimension of eigenspace $= 1$** **For $\lambda = 3$: Dimension of eigenspace $= 1$**
A	3	Find AM and GM for each of the eigenvalue of matrix $A = \begin{bmatrix} -2 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix}$. **Answer: For $\lambda = -2$: AM $= 2$ and GM $= 2$** **For $\lambda = 3$: AM $= 2$ and GM $= 1$**

B	4	Find the dimension of the eigenspace for each eigenvalues of the matrix $A = \begin{bmatrix} 19 & -9 & -6 \\ 25 & -11 & -9 \\ 17 & -9 & -4 \end{bmatrix}$. **Answer: For $\lambda = 1$: Dimension of eigenspace $=1$** **For $\lambda = 2$: Dimension of eigenspace $= 1$**
B	5	Find $\eta(A - \lambda I)$ for each λ of $A = \begin{bmatrix} 0 & 0 & 2 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ for $\lambda = 1$ and -1. **Answer: $\eta(A - I) = 2$, $\eta(A + I) = 1$**
C	6	Let $A_{4\times4}$ be a matrix with $\lvert A \rvert = 0$, $\text{tr}(A) = 0$ and the eigenvalues of the matrix A are $-3, 3, x$ and y then find the possible values of $\eta(A - \lambda I)$ for each λ. **Answer: $\eta(A) = 1$ or 2, $\eta(A - 3I) = 1$, $\eta(A + 3I) = 1$**
C	7	Let $A_{5\times5}$ be a matrix with $\lvert A \rvert = 41140$, $\text{tr}(A) = 60$ and the eigenvalues of the matrix A are $1, 11, 17, p$ and q then find the possible values of $\eta(A - pI)$ and $\eta(A - qI)$. **Answer: $\eta(A - pI) = \eta(A - 11I) = 1$ or 2, $\eta(A - qI) = \eta(A - 20I) = 1$**
B	8	For a matrix of order 3 has an eigenvalue $\lambda = 4$ whose eigenspace is spanned by the two linearly independent vectors. What can be the possible algebraic multiplicity of $\lambda = 4$? **Answer: 2 or 3**
B	9	If a non-zero matrix B has a characteristic polynomial of $\lambda^3(\lambda + 2) = 0$, then how many free variables will be present in the system $(B + 2I)X = 0$? **Answer: 1 free variable**

Method 4 ⇝ Diagonalization of Matrices

Examples of Method-4: Diagonalization of Matrices

B	1	Find a diagonal matrix D which is similar to the matrix $A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & -1 \\ 1 & 3 & -2 \end{bmatrix}$. **Answer:** $\begin{bmatrix} \mathbf{2} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{-1} \end{bmatrix}$
B	2	Find diagonalizing matrix and diagonal matrix for a given matrix $A = \begin{bmatrix} 1 & -6 & -4 \\ 0 & 4 & 2 \\ 0 & -6 & -3 \end{bmatrix}$. **Answer:** $\mathbf{P} = \begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{1} \\ \mathbf{0} & \mathbf{-\frac{2}{3}} & \mathbf{-\frac{1}{2}} \\ \mathbf{0} & \mathbf{1} & \mathbf{1} \end{bmatrix}$, $\mathbf{D} = \begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}$
B	3	Show that the matrix $A = \begin{bmatrix} -3 & 5 & -4 \\ -14 & 15 & -10 \\ -9 & 8 & -4 \end{bmatrix}$ is not diagonalizable. **Hint: It does not have n linearly independent eigen vector.**
A	4	Show that the matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 3 & 0 & 1 \\ 2 & 0 & 0 \end{bmatrix}$ is not diagonalizable by using the AM and GM of eigenvalues. **Hint: AM and GM are not same for some eigenvalue.**
A	5	Show that the matrix $A = \begin{bmatrix} 2 & 3 & 4 \\ 0 & 2 & -1 \\ 0 & 0 & 1 \end{bmatrix}$ is not diagonalizable by using the AM and GM of eigenvalues. **Hint: AM and GM are not same for some eigenvalue.**

B	6	Show that there is no diagonal matrix for matrix $A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$. **Hint: P^{-1} does not exist.**
C	7	Which of the following matrices are similar to diagonal matrix D? If yes, then find a diagonal matrix D, if no then give the reason. $A = \begin{bmatrix} 3 & -1 \\ 1 & 5 \end{bmatrix}$, $B = \begin{bmatrix} -17 & 18 & -6 \\ -18 & 19 & -6 \\ -9 & 9 & -2 \end{bmatrix}$, $C = \begin{bmatrix} 5 & -3 & 0 & 9 \\ 0 & 3 & 1 & -2 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$ **Answer: A is not similar to any diagonal matrix D,** **B is similar to diagonal matrix $D_B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}$ and** **C is similar to diagonal matrix $D_C = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}$.**
B	8	Determine which of the following matrices are diagonalizable? Give reason for each case. $A = \begin{bmatrix} 1 & -3 \\ -3 & 0 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 4 & -2 \\ -3 & 4 & 0 \\ -3 & 1 & 3 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 2 & 0 & 9 \\ 0 & 3 & 0 & -2 \\ 0 & 0 & 4 & 2 \\ 0 & 0 & 0 & 4 \end{bmatrix}$ **Answer: A is diagonalizable because it is real symmetric.** **B is diagonalizable because it has distinct eigenvalues.** **C is not diagonalizable beacause AM ≠ GM for some eigenvalue.**

C	9	Determine which of the following matrices are similar to diagonal matrix? Give reason for each case. $A = \begin{bmatrix} 2 & 8 & 2 \\ -1 & -4 & -1 \\ 3 & 12 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 16 & 4 \\ -2 & -9 & -2 \\ 6 & 24 & 5 \end{bmatrix}$, $C = \begin{bmatrix} -1 & 2 & -1 & 9 \\ 0 & 3 & 9 & -2 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 4 \end{bmatrix}$ **Answer: A is diagonalizable because it is idempotent matrix.** **B is diagonalizable because it is involutory matrix.** **C is diagonalizable beacause it has distinct eigenvalues.**

Method 5 ⇝ Quadratic Forms and Canonical Forms

Examples of Method-5: Quadratic Forms and Canonical Forms

A	1	Write down the Quadratic forms corresponding to the following matrices. (i) $A = \begin{bmatrix} 6 & 2 & -3 \\ 2 & 0 & 5 \\ -3 & 5 & 5 \end{bmatrix}$, (ii) $B = \begin{bmatrix} 1 & 2 & -4 & 0 \\ 2 & 5 & 1 & \frac{1}{2} \\ -4 & 1 & 0 & 9 \\ 0 & \frac{1}{2} & 9 & -6 \end{bmatrix}$ **Answer**: **(i) $6x^2 + 5z^2 + 4xy - 6xz + 10yz$** **(ii) $x^2 + 5y^2 - 6w^2 + 4xy - 8xz + 2yz + yw + 18zw$**
A	2	Express the quadratic form $Q(x) = x_1^2 + 2x_2^2 + 3x_3^2 + x_4^2 - 2x_1x_2 + 4x_1x_3 - 2x_1x_4 + 4x_2x_3 - 6x_2x_4 + 8x_3x_4$ in matrix notation. **Answer**: $\mathbf{x^TAx} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \begin{bmatrix} 1 & -1 & 2 & -1 \\ -1 & 2 & 2 & -3 \\ 2 & 2 & 3 & 4 \\ -1 & -3 & 4 & 1 \end{bmatrix} \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix}$

A	3	Express the following quadratic forms in matrix notation: (i) $4x^2 - 3xy + 2y^2$ (ii) $9x^2 + 3y^2 - 4z^2 + 8xy + 8xz - 7yz$ (iii) $x_1^2 - 2x_2^2 + 3x_3^2 + 7x_4^2 + 4x_1x_2 + 6x_1x_3 - x_1x_4 + 4x_2x_3 - 5x_2x_4 + 6x_3x_4$

Answer: **(i)** $\mathbf{x^T A x} = \begin{bmatrix} x \\ y \end{bmatrix} \begin{bmatrix} 4 & -\frac{3}{2} \\ -\frac{3}{2} & 2 \end{bmatrix} \begin{bmatrix} x & y \end{bmatrix}$

(ii) $\mathbf{x^T A x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \begin{bmatrix} 9 & 4 & 4 \\ 4 & 3 & -\frac{7}{2} \\ 4 & -\frac{7}{2} & -4 \end{bmatrix} \begin{bmatrix} x & y & z \end{bmatrix}$

(iii) $\mathbf{x^T A x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & -\frac{1}{2} \\ 2 & -2 & 2 & -\frac{5}{2} \\ 3 & 2 & 3 & 3 \\ -\frac{1}{2} & -\frac{5}{2} & 3 & 7 \end{bmatrix} \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix}$

A	4	Express the following quadratic forms in matrix notation: (i) $x^2 - 6xy + y^2$ (ii) $-x^2 - 2z^2 - 5xy + 4xz + 3yz$ (iii) $-x_1^2 + 3x_2^2 + 6x_3^2 + 8x_1x_3 + 2x_1x_2 + 6x_2x_3 - 9x_1x_4 - 5x_2x_4 + 12x_3x_4$ **Answer**: (i) $\mathbf{x^TAx} = \begin{bmatrix} x \\ y \end{bmatrix} \begin{bmatrix} 1 & -3 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} x & y \end{bmatrix}$ (ii) $\mathbf{x^TAx} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \begin{bmatrix} -1 & -\frac{5}{2} & 2 \\ -\frac{5}{2} & 0 & \frac{3}{2} \\ 2 & \frac{3}{2} & -2 \end{bmatrix} \begin{bmatrix} x & y & z \end{bmatrix}$ (iii) $\mathbf{x^TAx} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \begin{bmatrix} -1 & 1 & 4 & -\frac{9}{2} \\ 1 & 3 & 3 & -\frac{5}{2} \\ 4 & 3 & 6 & 6 \\ -\frac{9}{2} & -\frac{5}{2} & 6 & 0 \end{bmatrix} \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix}$
B	5	Find an orthogonal change of variable that eliminates the cross-product terms in the quadratic form $Q(x) = 2x_1^2 + 2x_2^2 + 2x_3^2 - 2x_1x_3$. **Answer**: $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{2}{3} & -\frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & \frac{2}{3} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} y_1 & y_2 & y_3 \end{bmatrix}$

B	6	Find an orthogonal change of variable that eliminates the cross-product terms in the quadratic form $Q(x) = 3x_1^2 + 4x_2^2 + 5x_3^2 + 4x_1x_2 - 4x_2x_3$. **Answer:** $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -\frac{2}{3} & \frac{2}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} & \frac{2}{3} \\ \frac{1}{3} & \frac{2}{3} & -\frac{2}{3} \end{bmatrix} \begin{bmatrix} y_1 & y_2 & y_3 \end{bmatrix}$
C	7	Reduce the following quadratic form $Q(x) = 2x^2 + 2y^2 - z^2 - 4yz + 4xz - 8xy$ to canonical form. **Answer: Canonical form $Q(x) = y^T(P^TAP)y = 7y_1^2 - 2y_2^2 - 2y_3^2$**
C	8	Reduce the following quadratic form $Q(x) = 2x_1^2 + 5x_2^2 + 5x_3^2 + 4x_1x_2 - 8x_2x_3 - 4x_3x_1$ to canonical form. **Answer: Canonical form $Q(x) = y^T(P^TAP)y = y_1^2 + y_2^2 + 10y_3^2$,**
B	9	Determine the rank, index, signature and value class (nature) of the following quadratic forms: (i) $6x_1^2 + 3x_2^2 + 3x_3^2 - 4x_1x_2 - 2x_2x_3 + 4x_3x_1$ (ii) $-3x_1^2 - 3x_2^2 - 3x_3^2 + 2x_2x_3 - 2x_3x_1 - 2x_1x_2$ **Answer: (i) $r = 3$, $p = 3$, $s = 3$, Positive definite** **(ii) $r = 3$, $p = 0$, $s = -3$, Negative definite**

B	10	Determine the value class of following quadratic forms by nature of eigenvalues. (i) $(x-y)^2$ (ii) x^2+y^2 (iii) $3x^2-y^2+2z^2+2xy+4xz+6yz$ (iv) $-4x^2-2y^2-2z^2-2xy+2xz-2yz$ (v) $-(x-y)^2-z^2$ **Answer: (i) Positive semidefinite, (ii) Positive definite, (iii) Indefinite, (iv) Negative definite, (v) Negative semidefinite.**
B	11	Determine the value class of following quadratic forms by nature of eigenvalues. (i) $(x+y)^2$, (ii) $-3x^2-2y^2-4z^2-2xy-2xz-2yz$ (iii) $3x^2+2y^2+3z^2-2xy-2yz$ (iv) $-5x^2-2y^2-10z^2+2xy-14xz+4yz$ **Answer: (i) Positive semidefinite, (ii) Negative definite, (iii) Positive Definite, (iv) Negative semidefinite.**
A	12	Determine the value class of following quadratic forms by nature of leading principal minors. (i) $x^2+2y^2+2z^2-2xy-2yz+zx$ (ii) $-4x^2-2y^2-2z^2-2xy+2xz-2yz$ **Answer: (i) Positive Definite, (ii) Negative deifinite**
B	13	Find the value of k for which the quadratic form $Q(x)=3x_1^2+x_2^2+2x_3^2-2x_1x_3+2kx_2x_3$ is positive definite. **Answer:** $-\sqrt{\frac{5}{3}} < k < \sqrt{\frac{5}{3}}$

B	14	Find the value of k so that the value class of the quadratic form $Q(x) = k(x^2 + y^2 + z^2) + 2xy - 2yz + 2zx$ is positive definite. **Answer: $k > 2$**
C	15	Find the values of k for which the quadratic form $Q(x) = -6x_1^2 + 3x_2^2 + kx_3^2 - 10x_1x_2 - 12x_1x_3 + 2kx_2x_3$ is negative definite. **Answer: No such k exists.**
C	16	Discuss the values of k for which the quadratic form $Q(x) = -7x_1^2 - x_2^2 + x_3^2 - 4x_1x_2 + 2kx_2x_3$ is neither positive definite nor negative definite. **Answer: For all $k \in \mathbb{R}$.**

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